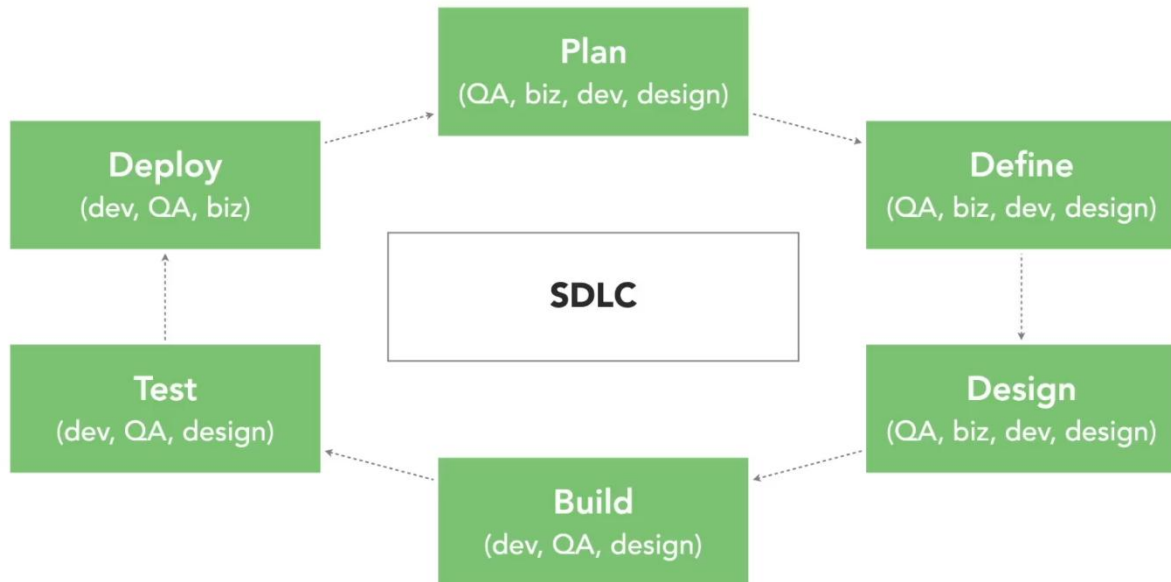


Software Testing

- SDLC – Software Development Life Cycle



- Test Plan ---
 1. Each Features should have associated with test plan.
 2. Test Plan can be described with Happy Path, Sad Path and Edge Case Scenarios.
 3. Test plan has 4 Steps: Scenario, Expected Result, Latest result and Automated

Example. Add or Delete New User

Scenario	Expected result	Latest result	Automated
Click a button to add a new user.	The create user modal opens and has two required fields for username and email. There is a create user button that becomes enabled once the fields are complete.	Pass	Yes

- **Acceptance Criteria**

1. Developer – Developer knows what to implement code for
2. Business Analyst – BA knows what scope the Story cover
3. QA – QA knows which scenario to test

Acceptance Criteria follow specific format

Given – precondition or beginning state of scenarios

When – input and action of scenarios

Then – Expected outcome of scenarios

Example User story

1) Add a sold out item to a Cart.

- Given I am viewing an item
- When I press the **Add to cart** button
- Then the item is added to the cart

2)Add an item to a cart (Error Scenario)

- Given I am viewing an item that is sold out
- When I press the **Add to cart** button
- Then the item is not added to the cart
- **And** I see a message that the item is out of stock

Done Specifies when Testing is complete

Sign off of Acceptance criteria from product, design and QA

Black Box Testing – Where you test an application without looking at its inner code or design.

You cannot see what is inside it. You only know what goes in(input) and what should come out(output).

Example – Vending Machine

1)The **Input** – You press the button for a bag of chips

2)Expected **Output** – A bag of chips drop down

3)The **Test** – You do not need to know how the gears inside read or how the computer chips inside read your button press.

You only care that when you press button, you get your chips.

Black Box Testing Types:

1)Functional Testing

2) Non-Functional testing

Gray Box Testing – Combination of Black Box testing + White Box Testing

White Box Testing

It is software testing method that examine the internal structure, Architecture and source code of an application.

Also known as clear box, glass box or here the box is completely transparent

And focus on the internals of the application and what is happening at the code or system level.

Example – Shopping Cart on an e-commerce web application

Items can be added and removed from the shopping cart and total is displayed based on the quantity of items in the cart.

Type of White Box Testing

1) Unit Testing

2) System Testing

Manual Testing ---

Identify the test scenario (Who is to be tested) before testing. Identify both typical (Happy Path) and nontypical (Negative Testing) use cases.

Test scenario Example: Verify the user login functionality.

Typical Use Cases (Happy Path): - Positive Testing

Valid Inputs and expected behaviours

Example of Use cases: An E-commerce checkout

- 1) Entering valid inputs like valid 16 digits credit card no, CVV and Expiry Date
- 2) Adding In stock item to the cart and proceeding through checkout successfully
- 3) Trying to register email address and correct password to log in.

Nontypical Use Cases (Edge cases): Negative Testing

Invalid input or non-standard user workflow

Example of Use cases: An E-commerce checkout

- 1) Entering special character or letters into credit card no field (Negative Input)
- 2) Setting the item quantity in the cart to 0, -1, or 999 (Boundary values)
- 3) Clicking the “submit order” button multiple time
- 4) Expiry data – trying to complete a purchase using expired coupon, expire credit card.

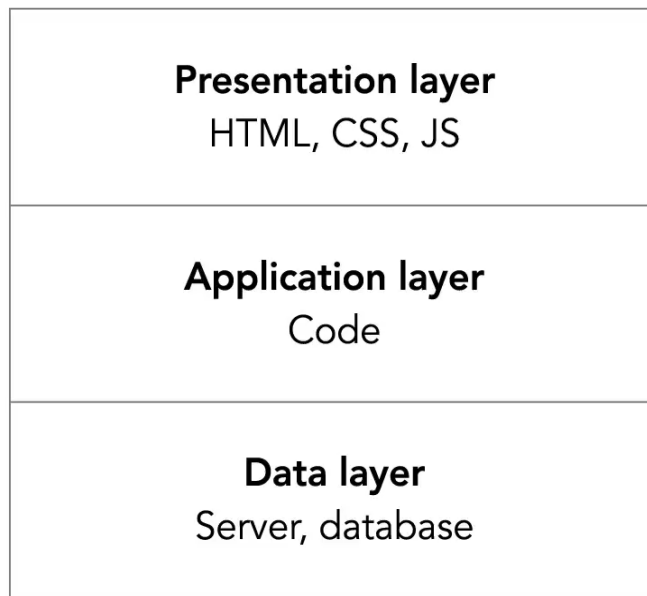
UI Automation:

Benefits – Scenario can be executed repeatedly and catch regression introduced in the application.

Integration testing

It focuses on the interaction between application components.

Its test is grey box testing.



Performance Testing

a broad software testing practice that evaluates how an application performs under various workloads.

example: an

E-commerce Platform (like Amazon) preparing for a massive **Black Friday sale**.

The Scenario

- **Normal daily traffic:** 10,000 active users at any given time.
- **Peak Black Friday prediction:** 100,000 concurrent users.

Stress Testing

is a type of non-functional testing that determines how an application behaves under extreme workloads. (Extreme Traffic)

Finds the breaking point and evaluates system error handling and recovery.

Goal of Stress Testing

- 1)When does it fail?
- 2)How does it recover?

Security Testing:

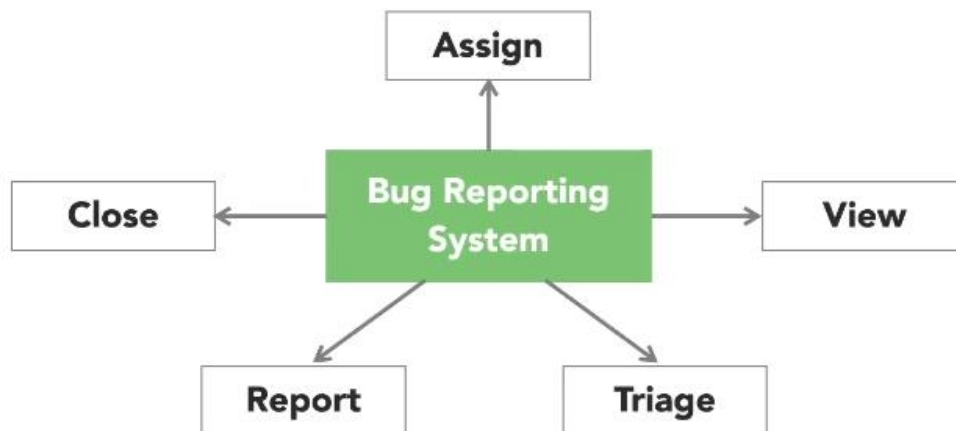
Security Testing exposes problem in the application that can cause unexpected behaviour and crashes.

Real World Example E-Commerce Application

- **SQL Injection (SQLi) Testing:** The tester inputs malicious code into the "Search" bar (e.g., ' OR '1'='1) to see if they can bypass the login screen or force the database to leak all user passwords.
- **Cross-Site Scripting (XSS) Testing:** The tester attempts to paste a malicious JavaScript script into a product review field. If the system is insecure, that script will execute in the browser of every other customer who views that product page, potentially stealing their session cookies.
- **Broken Object Level Authorization (BOLA) Testing:** A logged-in customer is viewing their profile page, and the URL reads ://shop.com. The tester manually changes the URL to ://shop.com to see if the system accidentally displays another customer's private address and order history.
- **Brute Force Testing:** An automated script rapidly inputs thousands of password combinations into the login portal to check if the platform enforces a rate limit (e.g., locking the account after 5 failed attempts).

Bug

Unexpected and Incorrect behaviour in an application.



Bug Report

Name:	Example
Description:	Items are not added to the cart after pressing the Add to cart button.
Steps to reproduce:	Select the first item and choose Add to cart .
Expected result:	When I go to the cart, I can see the item I selected has been added.
Actual result:	When I go to the cart, there are no items.
Screenshots:	
Browser/version:	Chrome 71
Logs:	
Tag:	Todd
Priority:	High
Status:	Ready for development

Triage Bug

Severity – How impactful the bug is to the business.

Priority – How Fast the bug should be fixed.

Types:

1) Low Severity Low Priority

Minimal customers

Don't affect major workflow

Example A small spelling mistake on an internal page
(Small problem + can fix later)

2) Low Severity High Priority

Not causing a lot of disruption to the site

Example Company name is misspelled on the homepage
before launch (Small problem + fix quickly)

3) High Severity Low Priority

Major problem only under some circumstances

Example

(Small problem + fix quickly)

4) High Severity High Priority

Major problems for many customers

Service disruption and unusual application in some workflow

Example Login button crashes the entire website (Big problem + fix immediately)

Severity	Priority	Example
High	High	Amazon website cannot place any orders
High	Low	A rarely used seller feature is not working

Severity	Priority	Example
Low	High	Amazon logo is displayed incorrectly on the homepage before a big launch
Low	Low	A small spelling mistake in the Amazon Help page